

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input Content was flagged HIGH priority in elicitation due to the deployment's requirement to handle moderately noisy hospital notes with abbreviations and shorthand. The benchmark provides standard-French biomedical text [\[Q19\]](#) across diverse sources, which superficially aligns with the deployment language. However, multiple critical concerns surface. First, no DrBenchmark dataset uses raw hospital EHR notes; sources are clinical trial protocols (ESSAI) [\[Q43\]](#), competition corpora (DEFT-2021, DEFT-2020) [\[Q24\]](#), peer-reviewed published case reports (CAS, CLISTER, E3C, DiaMed) [\[Q42, Q41, Q29, Q45\]](#), drug regulatory documents (QUAERO-EMEA) [\[Q34\]](#), MEDLINE titles, and patent claims. Web-source confirmation: no French raw-EHR clinical NLP benchmark exists [\[WEB-1, WEB-18\]](#). Examples of formality gap include CAS-D20, E3C-D9, DEFT2021-D14, DIAMED-D12. Second, substantial non-Metropolitan content contaminates several datasets: DiaMed's cases originate from Burkina Faso, Morocco, and Niger (DIAMED-D7, DIAMED-D8, DIAMED-D9); MorFITT contains Canadian, Jordanian, Saudi, and Arabic-population research content (MORFITT-D10, MORFITT-D20, MORFITT-D22, MORFITT-D31); E3C non-French configs (Basque, English, Italian, Spanish) dominate the multilingual structure (E3C-D7, E3C-D8). Third, DEFT-2020 contains 30–40% off-domain encyclopedic material (DEFT2020-D9, DEFT2020-D10, DEFT2020-D11). Fourth, PxCorpus — the closest noise proxy — contains system-prompt contamination corrupting clinical register (PXCORPUS-D22, PXCORPUS-D23). The CAS, CLISTER, DEFT-2021, and ESSAI corpora provide partial metropolitan-French clinical signal with quantified lab values and standard medical terminology (CAS-D2, CLISTER-D2, DEFT2021-D7), but all are edited published materials with cleaner register than active hospital notes (CAS-D20).

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q24	"DEFT-2021 (Grouin et al., 2021) is a subset of 275 clinical cases taken from the 2019 edition of DEFT." (p.3)
Q29	"E3C (Magnini et al., 2020) is a multilingual dataset of clinical cases annotated for the NER task." (p.3)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)

CAS-D2	“les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique...” — Dense lab-value notation typical of deployment's severity signal extraction
CLISTER-D2	Le bilan rénal a objectivé une insuffisance rénale avec une créatininémie à 1440 µmol/l, soit une clairance de la créatinine à 6,7 ml/mn/m2, urémie à 2,07 g/l. — Quantified biological workup with clinical abbreviations; representative of standard hospital documentation
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
E3C-D7	Azken hilabeteetan lkernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu. — Basque language — entirely irrelevant to French deployment context
DEFT2021-D7	bilan biologique d'entrée révélait une macrocytose avec une thrombopénie, un cytolyse majeure avec des ALAT et des ASAT respectivement à 60 à 450 fois la normale — Dense laboratory values with quantified severity signals
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
E3C-D8	Aurrekariak: - Krisi epileptikoa 12 urte zituenean, bere herrialdean tratatua: BOLIBIA. — Basque text referencing Bolivia — not relevant to French metropolitan hospital NLP
E3C-D9	Le bilan biologique montrait une cholestase (bilirubine totale a ` 140 mmol/L... — Well-formed published case report prose — contrasts with noisy active-hospital EHR shorthand
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
DEFT2020-D9	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaires datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure content; entirely off-domain for clinical NLP deployment
MORFITT-D10	L'objectif de cette étude était de documenter les pratiques actuelles de gestion des maladies dans les exploitations apicoles situées au sud-ouest du Québec, Canada. — Canadian beekeeping disease management study; non-Metropolitan French origin
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DEFT2020-D11	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Professional hockey franchise; irrelevant to hospital deployment
DIAMED-D12	L'examen clinique révélait une paralysie flasque aréflexique avec amyotrophie importante des membres inférieurs. On notait la présence de quelques trépidations épileptoïdes du pied gauche. — Polished academic prose — formal register unlike noisy active-hospital EHR notes
DEFT2021-D14	L' ECBU était stérile . — Standard abbreviation in complete sentence — published case report register, not raw EHR shorthand
CAS-D20	“vous êtes appelés au secours d' une infirmière de nuit pour confusion chez un patient bronchopathe chronique , tabagique non sevré à 70 paquets-années .” — Narrative clinical description; more formal than raw EHR shorthand
MORFITT-D20	79 patients au total ont été recrutés à Amman (Jordanie) en 2015. — Jordanian patient population; non-Metropolitan French clinical context
MORFITT-D22	le questionnaire YFAS 2.0-A... peut être utilisée comme équivalent du questionnaire anglais YFAS 2.0 pour étudier l'addiction à la nourriture dans les populations arabophones. — Arabic-population food addiction validation study; non-French population
PXCORPUS-D 22	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment teralithe 250 milligrammes 2 comprimés matin et soir pendant 1 mois — System prompt embedded in prescription utterance; corrupts register
PXCORPUS-D 23	le médicament mais pas le bon le système ne m'a pas compris le bon médicament et le 10 antalvic en gélules — User commentary about ASR failure; non-clinical register
MORFITT-D31	Un autoquestionnaire envoyé par courrier électronique a été rempli par 191 médecins internes sur 207 travaillant dans différentes spécialités et régions de Riyad. — Saudi Arabian intern survey; non-Metropolitan French context
WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)

Strengths

- CAS, CLISTER, and DEFT-2021 provide authentic French clinical case prose with quantified lab values, surgical narratives, and diagnostic reasoning closely matching standard hospital documentation register (CAS-D2, CLISTER-D2,

DEFT2021-D7)

- Explicit diversity of data origins — scientific literature, clinical trials, clinical cases, speech transcriptions [Q19] — provides cross-register robustness signal
- PxCorpus offers the benchmark's only naturally noisy register (transcribed speech with hesitations, truncations, ASR artifacts) (PXCORPUS-D1, PXCORPUS-D2, PXCORPUS-D4)

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
PXCORPUS-D1	euh 2 le matin 2 le midi et 2 le soir — Spoken filler in prescription; illustrates non-formal register
CAS-D2	"les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique..." — Dense lab-value notation typical of deployment's severity signal extraction
CLISTER-D2	Le bilan rénal a objectivé une insuffisance rénale avec une créatininémie à 1440 µmol/l, soit une clairance de la créatinine à 6,7 ml/mn/m2, urémie à 2,07 g/l. — Quantified biological workup with clinical abbreviations; representative of standard hospital documentation
PXCORPUS-D2	lamotrigine 25 milligrammes euh p/ combien — Incomplete utterance with filler and truncated word; spoken transcription noise
PXCORPUS-D4	ne pas tenir compte à midi tous les jours merd/ — Truncated word at end; raw transcription behavior
DEFT2021-D7	bilan biologique d'entrée révélait une macrocytose avec une thrombopénie, un cytolysse majeure avec des ALAT et des ASAT respectivement à 60 à 450 fois la normale — Dense laboratory values with quantified severity signals

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Inputs are in standard medical French and largely match the deployment's primary language. However, the deployment processes notes with French clinical abbreviations (ATCD, TTT, HDJ) and dictation artifacts that are absent or rare in benchmark sources. Region-specific PMSI/CIM-10 FR coding conventions [WEB-2] are not represented in the source text register.

IC-2: Substantial non-Metropolitan French clinical content is mixed into the benchmark. DiaMed cases are sub-Saharan/North African (DIAMED-D7, DIAMED-D8, DIAMED-D9); MorFITT contains Canadian (Quebec), Jordanian, Saudi, Arabic-population research (MORFITT-D10, MORFITT-D20, MORFITT-D22, MORFITT-D31); DEFT2021 contains some Quebec French terminology (DEFT2021-D20). This may distort performance estimates for the metropolitan target population.

IC-3: The benchmark does not contain Western-specific knowledge unsuitable to the metropolitan French context per se; however, DEFT-2020 contains substantial off-domain encyclopedic content (railways, video games, Russian historical figures, Japanese script) that introduces construct-irrelevant variance (DEFT2020-D9, DEFT2020-D10, DEFT2020-D12, DEFT2020-D13). MorFITT veterinary content is similarly off-domain (MORFITT-D5, MORFITT-D9).

IC-4: INSUFFICIENT DOCUMENTATION — would require recruitment of metropolitan French hospital clinicians and PMSI coders to assess whether benchmark content matches the moderate-noise register of active EHR notes.

IC-5: Critical content gaps: (1) no raw EHR notes [WEB-1, WEB-18]; (2) non-metropolitan content in DiaMed and MorFITT misrepresents the deployment population; (3) edited register systematically cleaner than active hospital documentation (CAS-D20, DEFT2021-D14, DIAMED-D12); (4) French clinical abbreviations (ATCD, TTT, HDJ) under-represented. These collectively harm content validity for the deployment.

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Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
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CAS-D2	"les constantes hémodynamiques étaient stables mais le bilan biologique mettait en évidence une anémie à 5 , 7 g / 100ml d' hb , une hyponatrémie à 128 meq / l , une acidose métabolique..." — Dense lab-value notation typical of deployment's severity signal extraction
MORFITT-D5	seules des données limitées sont disponibles sur le surfactant chez le cheval... notre objectif était de mener une étude immunocytochimique au microscope optique et électronique sur les poumons normaux du cheval. — Equine veterinary immunology study; off-domain for human hospital deployment
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
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MORFITT-D9	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe comparé aux chiens sains. — Canine ear microbiome study; veterinary content off-domain for hospital NLP
MORFITT-D10	L'objectif de cette étude était de documenter les pratiques actuelles de gestion des maladies dans les exploitations apicoles situées au sud-ouest du Québec, Canada. — Canadian beekeeping disease management study; non-Metropolitan French origin
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DEFT2020-D12	Lancement du célèbre MMORPG : World of Warcraft. — Video game launch event; entirely off-domain
DIAMED-D12	L'examen clinique révélait une paralysie flasque aréflexique avec amyotrophie importante des membres inférieurs. On notait la présence de quelques trépidations épileptoïdes du pied gauche. — Polished academic prose — formal register unlike noisy active-hospital EHR notes
DEFT2020-D13	42 Jeux indémodables (だれでもアソビ大全 , Daredemo Asobi Taizen au Japon...) — Japanese script embedded in French text; video game content; off-domain
DEFT2021-D14	L' ECU était stérile . — Standard abbreviation in complete sentence — published case report register, not raw EHR shorthand
MORFITT-D20	79 patients au total ont été recrutés à Amman (Jordanie) en 2015. — Jordanian patient population; non-Metropolitan French clinical context
DEFT2021-D20	Au jour 0, le jeune patient reçoit de l'acétaminophène 300 mg par voie orale — "Acétaminophène" (Quebec French) vs. "paracétamol" (metropolitan French) — minor regional lexical divergence
CAS-D20	"vous êtes appelés au secours d' une infirmière de nuit pour confusion chez un patient bronchopathe chronique , tabagique non sevré à 70 paquets-années ." — Narrative clinical description; more formal than raw EHR shorthand
MORFITT-D22	le questionnaire YFAS 2.0-A... peut être utilisée comme équivalent du questionnaire anglais YFAS 2.0 pour étudier l'addiction à la nourriture dans les populations arabophones. — Arabic-population food addiction validation study; non-French population
MORFITT-D31	Un autoquestionnaire envoyé par courrier électronique a été rempli par 191 médecins internes sur 207 travaillant dans différentes spécialités et régions de Riyad. — Saudi Arabian intern survey; non-Metropolitan French context

WEB-1	EHRNoteQA (NeurIPS 2024) demonstrates that existing clinical NLP benchmarks typically fail to represent raw multi-note EHR documentation; no equivalent French benchmark identified (https://arxiv.org/abs/2402.16040)
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)
WEB-18	Recent 2025 meta-analysis confirms a significant proportion of medical LLM benchmarks are constructed from academic materials rather than authentic clinical data — applicable to DrBenchmark's source composition (https://arxiv.org/html/2508.04325)

Information Gaps

- Quantitative measure of how performance on edited-register benchmarks transfers to noisy active EHR documentation in metropolitan French hospitals
- Full geographic distribution across all 739 DiaMed cases (only sample-confirmed as non-metropolitan)

Requires Expert Verification

- Whether French clinical abbreviation handling (ATCD, TTT, HDJ, etc.) by models trained on DrBenchmark is sufficient for deployment without a custom normalization layer

Remediation

Gap 1: No DrBenchmark dataset uses raw active-hospital EHR notes; benchmark sources are systematically cleaner than the deployment's moderately noisy hospital documentation, and DiaMed/MorFITT introduce non-metropolitan content (Burkina Faso, Niger, Morocco, Quebec, Jordan, Saudi Arabia)

Recommendation: Construct a deployment-internal validation set of metropolitan French hospital notes (with appropriate HDS-certified hosting [WEB-4]) annotated by hospital coders, sized for stratified evaluation across PMSI Catégories Majeures. Filter MorFITT and DiaMed evaluation results to exclude non-metropolitan cases when reporting deployment-relevant performance.

Gap 2: French clinical abbreviations (ATCD, TTT, HDJ, etc.) characteristic of hospital notes are under-represented in benchmark sources

Recommendation: Build a custom abbreviation expansion / normalization layer using French hospital coding documentation (CI-SIS / ANS standards) and validate it against held-out hospital notes before relying on DrBenchmark-trained models for production NER outputs.

ID	Evidence
WEB-4	https://engage.hoganlovells.com/knowledgeservices/news/health-data-hosting-the-new-french-hds-certification-has-been-released